

Agentic AI in Telecom & the Autonomous Network

After five days down in the physical infrastructure — silicon, protocols, power, sovereignty and cooling (Days 85–89) — Day 90 pivots back to where agents do the work, and lands on the industry that has to run them at carrier scale: the telco. The timing is not an accident. This week's DTW Ignite 2026 in Copenhagen (23–25 June) turned into a referendum on a single idea: the network should stop being automated and start being autonomous. The shift is from task-based automation — scripts that speed up predetermined steps while humans correlate and decide — to AI-native networks engineered to sense, decide and act on their own, with humans supervising rather than steering.

~38% CAGR growth of the AI-in-telecom market through 2030; 89% of operators say their AI budget will rise in the next 12 months, up from 65% a year ago	30–50% MTTR reduction reported from agentic network operations, alongside 20–25% OPEX savings as closed-loop control replaces manual correlation	40% fewer call-centre interactions where AI-native customer operations are deployed — automation has overtaken CX as the #1 investment use case	DTW Ignite '26 Copenhagen, 23–25 June: NVIDIA and partners rally telcos around trusted, 24/7 AI agents for autonomous network operations
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1 · From automation to autonomy — the AI-native network

Telcos have spent two years getting real returns from generative AI, but almost all of it has been task-based: automation that accelerates a known step while a human correlates the insight and directs what happens next. DTW Ignite 2026 marked the line being crossed. The new target is the autonomous network — a system that continuously senses operating conditions, decides, executes and then adapts, closing the loop without a human in the middle for routine work. The industry frames this as a maturity ladder (TM Forum's L0–L5 autonomous-network levels); most operators sit at partial automation today and are now reaching for conditional and high autonomy in bounded domains.

Two signals show the money following the thesis. The AI-in-telecom market is compounding at roughly 38% a year through 2030, and in this year's operator survey 89% said their AI budget will increase over the next 12 months — up from 65% a year ago. Just as telling, network automation has overtaken customer experience as the leading use case for new investment. The centre of gravity has moved from the call centre to the core.

Autonomy level	What the network does	Human role	Where telcos are
L1–L2 Assisted/Partial	Scripts + dashboards speed known steps	Operator drives every change	Mature / legacy baseline
L3 Conditional	Closed-loop optimisation in a domain	Approves, sets policy & guardrails	Live in RAN energy, congestion
L4 High autonomy	Agents detect, decide & act, cross-domain	Supervises exceptions only	2026 pilots (self-healing 5G)
L5 Full autonomy	Network self-runs end-to-end	Sets intent & outcomes	Aspiration / narrow slices

2 · The agentic network stack — a runtime you can trust

The hard part of carrier-grade agents was never intelligence; it was permission. An agent that can retune a radio or reroute traffic can also take down a region, so operators need behaviour that is predictable, auditable and governed before they grant access. That is exactly what the DTW announcements targeted. NVIDIA's blueprint pairs NeMo synthetic data (to train where real telemetry is sparse or sensitive), NemoClaw secure-runtime blueprints, and the OpenShell sandbox that gives agents policy-based, sandboxed access to production systems — every action logged, scoped and reversible.

Digital twins close the safety loop: RTX PRO 6000-accelerated replicas of the network let agents rehearse a change and observe its blast radius before it touches anything live. The pattern rhymes with the whole infrastructure series — trust, identity and governance are the gating capability, not the model. It is why the partner roster reads like an operating model, not a demo: SoftBank, Amdocs and NTT DATA are building on the same governed runtime so an agent's authority is bounded by policy, not by hope.

3 · Self-healing RAN & autonomous operations

Down at the radio, agents are already earning their keep. Closed-loop systems detect congestion patterns, tune RAN parameters, trigger small-cell activation, adjust power levels and resize network slices in response to live demand — the work that used to mean a 2 a.m. page and an hour of manual correlation. NTT DATA's long-running agents watch for network degradation and intervene before customers feel it; AdaptKey's security-hardened agents are piloting self-healing 5G operations. The reported envelope across early deployments — 30–50% lower mean-time-to-repair and 20–25% OPEX savings — comes precisely from removing the human from the routine detect-decide-act loop while keeping them on the exceptions.

The reframe for an operator is that reliability becomes a software property. A self-healing network is not one with fewer faults; it is one where the median fault is detected, diagnosed and resolved by an agent inside a guardrail, and only the genuinely novel reaches a human. Energy is the cleanest first domain: agents that sleep and wake cells against real traffic cut the single biggest controllable cost in the RAN without touching the customer experience.

4 · AI-native BSS/OSS, customer ops & the Telco-to-TechCo shift

Above the network, the same agentic pattern is rewriting the business stack. AI-native customer operations are deflecting up to 40% of call-centre interactions where deployed, while agents in BSS/OSS handle provisioning, order orchestration, billing disputes and field-service dispatch end-to-end. This is the operational core of the Telco-to-TechCo shift: a carrier that exposes its network as programmable, agent-callable capability — congestion, location, quality-on-demand via network APIs (the CAMARA effort) — stops selling only connectivity and starts selling outcomes that other people's agents can buy.

For an operator-side strategist the monetisation lens matters more than the cost-out one. Autonomous operations defend margin; agent-callable network APIs and AI-native customer journeys create new revenue. The winners treat the autonomous network not as a cheaper NOC but as the substrate of a platform business — which is precisely why the governance, identity and trust lessons from the infrastructure series are the unlock, not the friction.

Domain	What agents do	Signal / payoff
Self-healing RAN	Detect congestion, tune params, sleep/wake cells, slice	30–50% MTTR cut; OPEX -20–25%
Network ops / core	Proactive degradation detection, cross-domain change	NTT DATA, AdaptKey self-healing 5G
Customer operations	Care, retention, dispute resolution end-to-end	Up to 40% fewer call-centre touches
BSS/OSS	Provisioning, order orchestration, field dispatch	Faster fulfilment, lower error rates
Monetisation	Expose network as agent-callable APIs (CAMARA)	Telco-to-TechCo revenue, not just cost

Viral spotlight — NVIDIA's telecom agent stack — NemoClaw + OpenShell, the 'AI-native telco' goes governed

The week's viral infrastructure story in telecom was not a model but a runtime. At DTW Ignite, NVIDIA put a full agentic stack in front of operators: NeMo synthetic data to train where real network data is scarce, NemoClaw secure-runtime blueprints, the OpenShell sandbox for policy-based, sandboxed access to live telecom systems, and RTX PRO 6000-accelerated digital twins to rehearse changes before they touch the network. The pitch that travelled: long-running agents that are predictable, auditable and governed — guardrails first — so operators can actually let them act. Early movers gave it teeth. NTT DATA is using NVIDIA Nemotron open models with NemoClaw to build long-running agents that proactively detect network degradation; AdaptKey is piloting security-hardened agents for self-healing 5G operations; SoftBank and Amdocs are wiring agents across network and BSS. It is the cleanest proof of this issue's thesis: in telecom the breakthrough is not a smarter model, it is a runtime safe enough to give an agent the keys to a national network.

Guardrails-first NemoClaw secure runtime + OpenShell sandbox give agents policy-based, auditable, sandboxed access to live network systems	Digital twins RTX PRO 6000-accelerated twins let agents rehearse RAN/core changes before they touch production	NTT DATA · SoftBank · Amdocs early partners moving from task automation to long-running, self-healing network agents
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Market signal. After the physical-infrastructure arc (silicon, protocols, power, sovereignty, cooling), telecom is the first vertical to put governed, long-running agents on carrier-grade systems — and the constraint is exactly the one the infra series flagged: not model capability but a runtime safe enough to act. The market is voting (38% CAGR, 89% raising budgets, automation now the #1 use case), and the early payoff is concrete (30–50% MTTR, 20–25% OPEX, 40% fewer call-centre touches). The strategic prize, though, is not a cheaper NOC; it is the Telco-to-TechCo pivot, where agent-callable network APIs turn the autonomous network into a platform other agents pay to use. Track autonomy-ladder level and API exposure the way you track spectrum and coverage.

Practical takeaways

1. Start closed-loop on a bounded, measurable domain

Don't chase L5. Pick one domain where a number moves and the blast radius is contained — RAN energy (sleep/wake), congestion tuning, or tier-1 care deflection — and run a true closed loop there. MTTR and OPEX are the scoreboard; prove autonomy where you can measure it before you widen it.

2. Make the runtime, not the model, your day-one decision

Carrier-grade means an agent's authority is bounded by policy. Treat a NemoClaw/OpenShell-style secure runtime — sandboxed access, full audit, digital-twin rehearsal — as the gating requirement. If you can't sandbox, log and reverse an agent's action, it doesn't get the keys to the network.

3. Frame it as Telco-to-TechCo, not cost-out

Autonomous operations defend margin; agent-callable network APIs (CAMARA) create revenue. Put the monetisation lens on the roadmap from the start — expose congestion, location and quality-on-demand as capabilities other agents can buy, and the autonomous network becomes a platform, not just a cheaper NOC.

Tomorrow — Day 91: Agentic AI in Healthcare & Life Sciences — from the autonomous network to the clinic. Microsoft and Mayo Clinic's new provider-owned frontier model, the ambient-scribe beachhead (Abridge, Nuance DAX, the VA going nationwide), Hippocratic AI's 115M patient interactions, and why healthcare's higher safety and governance bar is the product, not the friction.